

Simplify Algebraic Expressions

Lesson 6-7

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

$5x$ and $3x$ are like terms (both x); $5x$ and $5y$ are NOT (different variables); $2x$ and $2x^2$ are NOT (different powers)

Like Terms

Terms with the same letter, like $2x$ and $5x$.

$8m + 3m = 11m$ — add the coefficients ($8 + 3 = 11$) and keep the variable (m)

Combine

To add or subtract terms with the same letter into one.

$2a + 5a + 3$ simplifies to $7a + 3$ — the two a -terms combine, the constant stays

Simplify

To write an expression in its shortest form.

In $9x$, the coefficient is 9 — it tells you 'nine groups of x '; if $x = 2$, then $9x = 18$

Coefficient

The number in front of a letter, like the 3 in $3x$.

Key Ideas & Notes

- The music studio is placing supply orders for three rehearsal rooms.
- Room A needs 4 microphones and 3 stands.
- Room B needs 2 microphones and 5 stands.
- Room C needs 6 microphones and 1 stand.
- Identify the like terms in each expression. Group the terms that can be combined.

Think About It

- Why can you add the microphones together but not mix microphones with stands?
- What makes microphones and stands 'different types' of items?
- How is this like combining like terms in algebra?

My Notes

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Three rooms need microphones and stands. Why can you add all the microphones together ($4 + 2 + 6$) but not mix microphones with stands?

Level 1 support

Start here: You can only combine items of the same type, just like like terms.

I can add the microphones because they are the ____ type.

Puedo sumar los micrófonos porque son del ____ tipo.

I cannot mix microphones and stands because they ____.

No puedo mezclar micrófonos y soportes porque ____.

Word bank: Like Terms Combine Simplify Coefficient

Listen for: Students explain microphones combine with microphones ($4 + 2 + 6 = 12$) but not with stands, connecting this to combining like terms in algebra.

Level 2

How is mixing microphones and stands like trying to add $3x$ and 5 ? Generalize the rule for which terms can combine.

- Mixing microphones and stands is like adding $3x$ and 5 because ____.
- In general, terms combine only when they have the ____.

EXPLORE

To simplify $6x + 4 + 2x + 7$, which are the like terms, and how do you combine them?

Level 1 support

Start here: Combine the x-terms by adding coefficients; combine the constants separately.

The like terms are ____ **and** ____.

Los términos semejantes son ____ y ____.

I combine them by ____ **to get** ____.

Los combino al ____ para obtener ____.

Word bank: **Like Terms** **Combine** **Simplify** **Coefficient**

Listen for: A strong answer combines $6x + 2x = 8x$ and $4 + 7 = 11$ to get $8x + 11$, adding coefficients of like terms and keeping the variable.

Level 2

A classmate simplifies $5x + 3 + 2y + 4x$ as $11xy + 3$. Critique the error and give the correct simplified form.

- The error is ____ because $5x$ and $2y$ are not ____.
- The correct simplified form is ____ because I combine only ____.

CONNECT

The festival order is $(8m + 5) + (3m + 7) + (4m + 2)$, where m is microphones per stage. What does the simplified expression tell you?

Level 1 support

Start here: Combine the m -terms and the constants to read the total at a glance.

The total simplifies to ____ m + ____.

El total se simplifica a ____ m + ____.

The ____ m tells us ____ and the constant tells us ____.

El ____ m nos dice ____ y la constante nos dice ____.

Word bank: Like Terms Combine Simplify Coefficient

Listen for: Students simplify to $15m + 14$, explaining $15m$ is the total cables tied to microphones and 14 is the extra fixed cables across stages.

Level 2

Why is $15m + 14$ easier to use for ordering than the original three grouped expressions? Justify how simplifying makes the real problem clearer.

- The simplified form is easier because ____.
- If each stage had 6 microphones, the total cables would be ____, which is faster to find because ____.

Guided Examples

Example 1

Simplify $7x + 3x$.

Solution: $7x + 3x = 10x$. Add the coefficients: $7 + 3 = 10$.

Answer: A. $10x$

Example 2

Simplify $9n + 4 + 2n$.

Solution: $9n + 2n = 11n$. The constant 4 has no like term, so it stays. Result: $11n + 4$.

Answer: A. $11n + 4$

Example 3

Which pair are like terms?

Solution: $6x$ and $2x$ are like terms because they both have the same variable (x) raised to the same power (1).

Answer: A. $6x$ and $2x$

Write About the Math

The Writing Revolution

I can explain my steps using the words like terms, combine, simplify, and coefficient.

1. Kernel Sentence subject + verb

Model: Simplify is to write an expression in its shortest form.

Simplificar es escribir una expresión en su forma más corta.

Write a kernel sentence about simplify. Use a subject and a verb.

Escribe una oración base sobre simplificar. Usa un sujeto y un verbo.

2. Sentence Expansion because · but · so

Kernel: Simplify matters in math

Simplificar importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Simplify matters in math because ____.

Simplificar importa en matemáticas porque ____.

but
pero

Simplify matters in math, but ____.

Simplificar importa en matemáticas, pero ____.

so
entonces

Simplify matters in math, so ____.

Simplificar importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement
Afirmación

Tell one true fact about simplify.
Di un hecho verdadero sobre simplify.

Simplify ____.

Question
Pregunta

Ask a question about simplify.
Haz una pregunta sobre simplify.

How does ____ ?

¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about simplify.
Muestra entusiasmo sobre simplify.

Wow, ____ !

¡Guau, ____ !

Command
Mandato

Tell a partner what to do with simplify.
Dile a un compañero qué hacer con simplify.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

I can add the microphones because they are the ____ type.

Puedo sumar los micrófonos porque son del ____ tipo.

I cannot mix microphones and stands because they ____.

No puedo mezclar micrófonos y soportes porque ____.

The like terms are ____ and ____.

Los términos semejantes son ____ y ____.

Try It

Solve on your own. Check the answer key when you are done.

1. Which pair are like terms?

- A. $6x$ and $2x$
- B. $6x$ and $6y$
- C. $6x$ and 6
- D. $6x$ and $2x^2$

Show your work:

2. Simplify $5a + 3 + 2a$.

- A. $7a + 3$
- B. $10a$
- C. $7a^2 + 3$
- D. $52a + 3$

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Three students simplified $4x + 3 + 2x + 5 + x$ differently. Student A got $7x + 8$. Student B got $7x^2 + 8$. Student C got $6x + 9$. Who is correct? Explain the mistakes the other two students made.

Sentence starter: Student ____ is correct because $4x + 2x + x = \underline{\hspace{1cm}}x$ and $3 + 5 = \underline{\hspace{1cm}}$. Student ____'s error was _____. Student ____'s error was _____.

Show your work:

Reflect — Exit Ticket

Simplify $6x + 3 + 2x + 5$.

- A. $8x + 8$
- B. $8x^2 + 8$
- C. $62x + 35$
- D. $16x$

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** A. $6x$ and $2x$ — *$6x$ and $2x$ are like terms because they both have the same variable (x) raised to the same power (1).*
2. **Try It 2:** A. $7a + 3 - 5a + 2a = 7a$. *The constant 3 stays. Result: $7a + 3$.*
3. **Exit Ticket:** A. $8x + 8 - 6x + 2x = 8x$ and $3 + 5 = 8$, so the simplified expression is $8x + 8$.

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Simplify is to write an expression in its shortest form.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).